

OPOYO: Ambient Floor Sensing for the Long Lie, without a Wearable or a Camera

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Abstract

A fall becomes an emergency when the person cannot get up and cannot call. That interval is a long lie. Buttons, pendants, and cameras fail the people who live alone and refuse to be watched. OPOYO is an ambient detector bonded to the floor slab. A contact sensor reads vibration. A microphone may suppress an alarm the floor model proposed. A confirmation window then waits to see whether anyone moves. Only a still, unblocked, person-like impact becomes an alert.

This paper is a CHI-shaped technical report of the design and of a phone-proxy proof of concept. The production intent classifies on the device and transmits an event record. The demo streams 50 Hz phone acceleration to a laptop that runs the same decision sequence. Envelope features (crest, rise, decay) are the live classifier. Spectral claims that need hundreds of hertz of bandwidth are out of reach of that phone stream. We report a collection protocol for testing those features on a given floor. We do not report precision, recall, or false alarms per month. Those rates are unmeasured.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; *Ubiquitous and mobile computing systems and tools*.

Keywords

fall detection, aging in place, ambient sensing, floor vibration, long lie, false alarms

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1 Introduction

Falls at home are common in later life, and many of the people who fall cannot rise unaided. Tinetti et al. measured that share at 47% among non-injured fallers [14]. Fleming and Brayne followed people over 90 and found that 80% of fallers could not get up [2]. In that cohort, 82% of falls happened while the person was alone. Thirty percent of those falls ended in an hour or more on the floor.

The clinical object is that hour. A long lie is an inability to get up for 60 minutes or more [7]. Dehydration, pressure injury, rhabdomyolysis, hypothermia, and aspiration accumulate while the person is

still conscious. Wild, Nayak and Isaacs associated lies longer than an hour with high six-month mortality [16]. Later reviews treat the causal link as unsettled. We cite the association, not a proven mechanism. In one Singapore emergency series, 85.3% of 720 elderly injury presentations were falls at home [17].

Singapore's deployed safety net is a wall button. The wireless Alert Alarm System reaches about 26,800 seniors across about 170 HDB rental blocks [9]. The button requires consciousness, reach, and willingness to press. Wearable pendants fail on a different axis: they are removed, left uncharged, or refused, and they are off in the shower and in bed, where falls concentrate [5, 6]. Cameras detect falls. Households reject them in the bedroom and bathroom.

False alarms decide whether any ambient detector stays installed. A device that pings for a saucepan is unplugged. Precision here is a retention requirement.

OPOYO is a sealed node bonded to the floor of a senior who lives alone in an HDB flat, still walks unaided, and refuses visible monitoring. The buyer is an adult child or an Active Ageing Centre. The user and the buyer want different things. The user wants coverage without a camera and without a thing to wear. The buyer wants an alert when a long lie is underway.

The design reads two channels of one physical event. Vibration on the slab carries the impact. Sound in the air can contradict a false impact. A confirmation window then asks whether the floor stays still. An alert is a person-like impact, unblocked by the air check, with no recovery in that window.

This paper's contributions are:

- (1) A problem framing for living-alone HDB seniors in which the long lie and false-alarm abandonment are the HCI constraints, not afterthoughts.
- (2) A sequential pipeline: energy onset, shape classification, acoustic suppression, confirmation, escalation.
- (3) A phone-proxy implementation of that pipeline, and a labelled-collection protocol for testing envelope features on one floor.
- (4) An explicit ledger of which numbers are physics, which are Alwan's measured range, which are runbook heuristics, and which are unmeasured.

Gait trending over months is a production extension. It is out of the proof of concept.

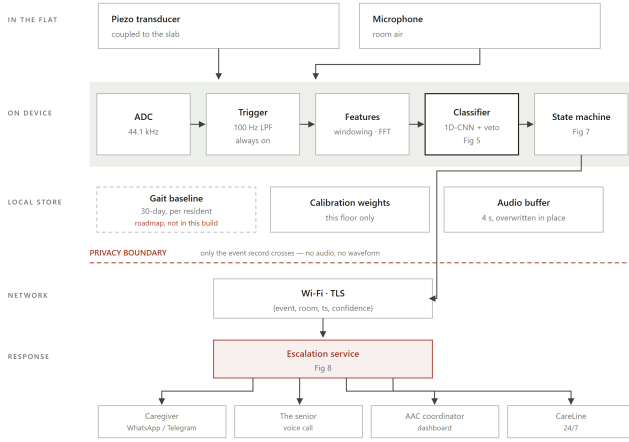


Figure 1: Production and demo share the event object and the escalation path. The phone stands in for the piezo during the build. Classification in production is on the node.

2 Related work

Fall detection splits into worn sensors, cameras, and ambient sensors [6, 12]. Each family solves detection and fails a different part of this segment.

Wearable systems use accelerometers on the wrist, waist, or pendant. They work when they are worn. Hawley-Hague et al. reviewed older adults' views of falls technologies and found stigma, charging, and removal as recurring barriers [5]. The shower and the night are exactly when those barriers bite.

Vision systems classify posture or motion from video. They work in line of sight. They fail the rooms where people undress. Privacy is the rejection, not a secondary preference.

Ambient systems use PIR, radar, Wi-Fi channel state, or the floor itself. Alwan et al. bonded a piezo to concrete and classified dummy falls against dropped objects [1]. Their useful range on a mezzanine concrete floor with linoleum was 20 ft, and 15 ft on a slab. We treat those figures as Alwan range, one construction, dummy falls. Litvak, Zigel and Gannot combined floor vibration with sound [8]. Minvielle et al. and Truong et al. published smart-floor datasets with under-floor piezo elements [11, 15]. Those corpora use sensors in the flooring build-up. They carry no paired audio. We use them as a reference for the vibration branch, not as evidence that a surface-bonded puck on HDB tile will match their scores.

HCI work on aging in place has documented the social job of not appearing to need watching. A pendant is a visible mark. A camera is a watch. A floor node has no affordance to press and no lens. That absence is the design, not an omission.

What prior ambient work leaves open for this segment is the combination of (i) a confirmation window aimed at the long lie, (ii) an acoustic check that may delete an alarm without requiring the fallen person to speak, and (iii) on-device classification so raw audio never leaves the flat. OPOYO is that combination. The demo has not yet measured whether it holds on a household floor.

3 Design goals

The user is 65 or older, lives alone in an HDB flat, walks unaided, and refuses visible monitoring. Singapore counted 88,400 residents aged 65 and over living alone in 2025 [13]. The old-age support ratio fell from 6.0 working-age adults per senior in 2014 to 3.5 in 2024, and is projected to 2.7 by 2030 [10]. Informal next-room noticing is the resource that is shrinking.

We designed against four goals, in this order.

G1. Coverage without action. Detection must hold in the bathroom at 3 am, with the door closed, with nothing worn.

G2. A false alarm the household will tolerate. A dropped pan, a slammed door, and a television must be able to fail the alarm path.

G3. Detect the long lie. A heavy sit and a fall can share a first second. The system waits for recovery motion before it pages.

G4. Nothing to watch. Raw audio and vibration stay on the node. The network sees an event record.

Gait as a precursor [4] is a fifth goal for a later build. The proof of concept labels fall or no fall only.

4 System

4.1 Sensing

Interior partition walls in an HDB flat stand on one continuous reinforced slab. A contact transducer bonded to that slab observes rooms that share it, including through a closed door. Airborne sound couples poorly into concrete, which is why a television can saturate a microphone and leave the slab quiet.

Coverage is finite. Alwan et al. measured about 4.6 m of useful radius on concrete [1]. A larger flat can need a second node. An expansion joint at the party wall reflects the wave, which is also why a neighbour's fall should not register. Which rooms a node covers is measured at install.

Peak amplitude does not separate a body from a rigid object. Duration of the decay and the spectrum of the impulse are the candidate cues. Our figure of a body decaying over about 1 s and a pan stopping within about 150 ms is a schematic caption. Those times are unmeasured on our floors.

4.2 Decision sequence

Let $x[n]$ be floor acceleration magnitude and $a[n]$ the matching air clip. The pipeline is five maps.

- (1) **Onset.** A 2 Hz high-pass removes gravity and slow drift. A 100 ms RMS is compared to $\theta = \text{median}(R) + k \cdot \text{MAD}(R)$, with $k = 6$ a runbook heuristic. Quiet frames alone update R . A hard floor of 3 m/s^2 (runbook heuristic) sits under θ . This stage asks whether something hit.
- (2) **Window.** The tail exists after the peak. The extractor waits 1.5 s, then copies 0.5 s before the hit and 1.5 s after it. The clip is 2 s so the decay sits inside it.
- (3) **Score.** Seven envelope and spectral numbers, or a 1-D CNN on the same clip, return $p \in [0, 1]$. The live demo uses the seven-number rule. The cut $p^* = 0.60$ is a placeholder until a precision-recall sweep exists.

- (4) **Air check.** A second classifier may delete the alarm. Fail-open: a broken check returns allow. YAMNet [3] wants 16 kHz PCM. The demo phone sends one dB per tick, so the live check is a loudness envelope behind the same interface. The fallen person is not required to speak. Silence is an allowed clip. Sustained loudness is a blocked clip.
- (5) **Confirm.** The demo waits 10 s. The product wait is 60 s. Motion yields recovered. Operator cancel yields cancelled. Silence yields alert, once.

Only alert may leave the machine. The payload is event id, room, time, p , audio label, peak, decay. Telegram rung 1 sends that payload. Rung 2 waits 60 s unless acknowledged. In production the same object is all that should cross the network.

4.3 Calibration

Floor construction dominates damping. A model trained on one tile will not keep its output layer on another. Install calibration records heel-drops as positives and dropped objects as negatives, embeds them with a frozen extractor, and fits the last layer. That is ordinary transfer. Fitting the whole net on ten drops would overwrite the extractor.

5 Implementation

The production node is a piezo, a MEMS microphone, and an on-device classifier. The hackathon build substitutes an iPhone lying on tile. Figure 1 is the shared contract: both rows emit the same event JSON. The phone samples user-acceleration at 50 Hz and a microphone RMS as dB. Packets go by UDP JSON to a laptop on port 9000. The laptop runs the five maps, a dashboard, and dry-run Telegram.

Nyquist at 50 Hz is 25 Hz. Spectral features that need hundreds of hertz of bandwidth cannot be observed on that stream. Crest, rise, and decay still can. Whether they separate a bag from a book on a given tile is an empirical question for the collection protocol below.

The dashboard shows a large state word, an RMS trace with the live threshold, and an event log. Cancel and acknowledge sit on that page. A standalone recorder writes labelled CSVs without the dashboard. Recorder and dashboard both bind UDP 9000, so they run in sequence.

Unit tests drive the shipped functions: high-pass of a DC signal, median-plus-MAD trigger, seven features, rule scores, fail-open air check, and a 10 s confirm that emits alert once. Those tests check the code path. They are not a field evaluation.

6 Evaluation plan

We will not invent a confusion matrix. The protocol is the contribution of this section.

Clips. One physical event is one file. The columns are timestamp and the three axes, plus magnitude and dB. Positives are heel-drop (ten takes) and an 8–12 kg bag dropped flat from about waist height (ten takes). A jump leaves the floor, so it is a different signal. Negatives are book, pan, door, walk, and quiet. Metadata per file: floor material, room, distance (fixed at 1.5–2 m), case on or off, object. The friend who records uses a standalone UDP writer. The dashboard is optional.

Split. Leave-one-file-out. Two windows cut from one drop are not independent.

Metrics. Precision, recall, F1, and a confusion matrix. Accuracy is discarded: a constant “no” looks good when falls are rare. The operating point is high recall with a cap on alerts from quiet and walk. False alarms per sensor-month is the retention number. It is unmeasured.

What would falsify the live features. If the decay_ms histograms of heel-drop and book overlap on this floor, the shape bet fails here. The honest report is that overlap.

7 Discussion

The HCI claim is the placement of the sensor. Putting detection in the slab is how G1 and G4 are met at once. The senior has nothing to charge. The bathroom stays covered. There is no image.

The retention claim is G2. Vibration onset is cheap and noisy. The air check and the confirmation window are the two filters that stand between an onset and a page. We have not shown that those filters bring false alarms below one per month.

The clinical claim is G3. Waiting 10 s is a demo value so a judge can watch the state change. Waiting 60 s is the product value. Neither is a long lie. They are a bound on time-to-alert after a still, person-like impact.

The demo streams raw axes to a laptop. That stream is a prototype convenience. It is the opposite of G4. Swapping the phone for a bonded puck is a change of the first stage only if both paths emit the same event object. That substitution is untested.

8 Limitations

The phone does not measure the slab. Absolute sensitivity on tile is not the sensitivity of a bonded piezo.

Fifty hertz cannot carry the paper’s frequency story about pans versus bodies.

$k = 6$, $p^* = 0.60$, and the 1.5 s wait are runbook heuristics and a placeholder. They are set from labelled files after histograms exist.

Drilling, a slow slump onto a mattress, and a fall outside Alwan range are unhandled miss or false-alarm modes.

No older adult has used the system. No household has lived with it for a month. Stigma, trust, and abandonment remain literature about other devices [5], not measurements of this one.

9 Conclusion

OPOYO is a sequential decision on floor vibration and air, built for a person who will not wear a pendant and will not accept a camera. The proof of concept runs that decision on a phone stream and a laptop. The next measurement is whether crest, rise, and decay separate heel-drop from book on the floor where the demo will stand. Until that measurement exists, the pipeline is a design and a tested program, not a fielded detector.

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